Welcome sales dat.py 2

Users > kartiksinghtomer > sales dat.py > ...

```python
import pandas as pd
import numpy as np
from datetime import datetime, timedelta

num_rows = 1000
categories = ['Electronics', 'Home Decor', 'Apparel', 'Office Supplies']
regions = ['North', 'South', 'East', 'West']

data = {
    'OrderID': range(1001, 1001 + num_rows),
    'Date': [datetime(2025, 1, 1) + timedelta(days=np.random.randint(0, 365)) for _ in range(num_rows)],
    'Category': [np.random.choice(categories) for _ in range(num_rows)],
    'Region': [np.random.choice(regions) for _ in range(num_rows)],
    'Quantity': np.random.randint(1, 10, size=num_rows),
    'UnitPrice': np.random.uniform(20.0, 500.0, size=num_rows).round(2)
}

df = pd.DataFrame(data)
df['TotalRevenue'] = df['Quantity'] * df['UnitPrice']
df.to_csv('sales_data.csv', index=False)
print("File 'sales_data.csv' created successfully.")
```

Welcome sales data dashboard.py 4 ●

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```python
import dash
from dash import dcc, html, Input, Output
import pandas as pd
import plotly.express as px

df = pd.read_csv('sales_data.csv')
df['Date'] = pd.to_datetime(df['Date'])

app = dash.Dash(__name__)

app.layout = html.Div(style={'fontFamily': 'Arial, sans-serif', 'padding': '20px'}, children=[
    html.H1("Sales & Revenue Analysis Dashboard", style={'textAlign': 'center'}),

    html.Div([
        html.Div([
            html.Label("Select Category:"),
            dcc.Dropdown(
                id='category-filter',
                options=[{'label': i, 'value': i} for i in df['Category'].unique()],
                multi=True,
                placeholder="All Categories"
            ),
        ], style={'width': '48%', 'display': 'inline-block'}),

        html.Div([
            html.Label("Select Date Range:"),
            dcc.DatePickerRange(
                id='date-filter',
                min_date_allowed=df['Date'].min(),
                max_date_allowed=df['Date'].max(),
                start_date=df['Date'].min(),
                end_date=df['Date'].max()
            ),
        ], style={'width': '48%', 'display': 'inline-block', 'float': 'right'}),
    ], style={'padding': '20px', 'backgroundColor': '#f9f9f9', 'borderRadius': '10px'}),

    html.Div(id='kpi-container', style={'display': 'flex', 'justifyContent': 'space-around', 'margin': '20px 0'}),
```

Welcome sales data dashboard.py 4

Users > kartiksinghtomer > sales data dashboard.py > ...

```python
    ], style={'padding': '20px', 'backgroundColor': '#f9f9f9', 'borderRadius': '10px'}),

    html.Div(id='kpi-container', style={'display': 'flex', 'justifyContent': 'space-around', 'margin': '20px 0'}),

    html.Div([
        dcc.Graph(id='revenue-trend-chart', style={'width': '65%', 'display': 'inline-block'}),
        dcc.Graph(id='category-pie-chart', style={'width': '33%', 'display': 'inline-block', 'float': 'right'}),
    ]),

    html.Div([
        dcc.Graph(id='region-bar-chart'),
    ])
])

@app.callback(
    [Output('kpi-container', 'children'),
     Output('revenue-trend-chart', 'figure'),
     Output('category-pie-chart', 'figure'),
     Output('region-bar-chart', 'figure')],
    [Input('category-filter', 'value'),
     Input('date-filter', 'start_date'),
     Input('date-filter', 'end_date')]
)
def update_dashboard(selected_categories, start_date, end_date):
    filtered_df = df[(df['Date'] >= start_date) & (df['Date'] <= end_date)]
    if selected_categories:
        filtered_df = filtered_df[filtered_df['Category'].isin(selected_categories)]

    total_rev = f"${filtered_df['TotalRevenue'].sum():,.2f}"
    total_qty = f"{filtered_df['Quantity'].sum():,}"
    avg_order = f"${filtered_df['TotalRevenue'].mean():,.2f}"

    kpis = [
        html.Div([html.H3("Total Revenue"), html.P(total_rev)], class_name='kpi-box'),
        html.Div([html.H3("Units Sold"), html.P(total_qty)], class_name='kpi-box'),
        html.Div([html.H3("Avg Order Value"), html.P(avg_order)], class_name='kpi-box'),
    ]
```

Welcome sales data dashboard.py 4 ●

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```python
def update_dashboard(selected_categories, start_date, end_date):
    total_qty = f"{filtered_df['Quantity'].sum():,}"
    avg_order = f"${filtered_df['TotalRevenue'].mean():,.2f}"

    kpis = [
        html.Div([html.H3("Total Revenue"), html.P(total_rev)], className='kpi-box'),
        html.Div([html.H3("Units Sold"), html.P(total_qty)], className='kpi-box'),
        html.Div([html.H3("Avg Order Value"), html.P(avg_order)], className='kpi-box'),
    ]

    trend_df = filtered_df.groupby('Date')['TotalRevenue'].sum().reset_index()
    fig_trend = px.line(trend_df, x='Date', y='TotalRevenue', title='Daily Revenue Trend')
    fig_pie = px.pie(filtered_df, names='Category', values='TotalRevenue', title='Revenue by Category', hole=0.4)
    region_df = filtered_df.groupby('Region')['TotalRevenue'].sum().reset_index()
    fig_region = px.bar(region_df, x='Region', y='TotalRevenue', title='Revenue by Region', color='Region')

    return kpis, fig_trend, fig_pie, fig_region

if __name__ == '__main__':
    app.run_server(debug=True)
```